
Contents

<i>Preface</i>	ix
1 Introduction	1
2 Variation	15
2.1 Statistics and Sampling Variation	15
2.2 Convergence	28
2.3 Order Statistics	37
2.4 Moments and Cumulants	44
2.5 Bibliographic Notes	48
2.6 Problems	49
3 Uncertainty	52
3.1 Confidence Intervals	52
3.2 Normal Model	62
3.3 Simulation	77
3.4 Bibliographic Notes	90
3.5 Problems	90
4 Likelihood	94
4.1 Likelihood	94
4.2 Summaries	101
4.3 Information	109
4.4 Maximum Likelihood Estimator	115
4.5 Likelihood Ratio Statistic	126
4.6 Non-Regular Models	140

4.7	Model Selection	150
4.8	Bibliographic Notes	156
4.9	Problems	156
5	Models	161
5.1	Straight-Line Regression	161
5.2	Exponential Family Models	166
5.3	Group Transformation Models	183
5.4	Survival Data	188
5.5	Missing Data	203
5.6	Bibliographic Notes	218
5.7	Problems	219
6	Stochastic Models	225
6.1	Markov Chains	225
6.2	Markov Random Fields	244
6.3	Multivariate Normal Data	255
6.4	Time Series	266
6.5	Point Processes	274
6.6	Bibliographic Notes	292
6.7	Problems	293
7	Estimation and Hypothesis Testing	300
7.1	Estimation	300
7.2	Estimating Functions	315
7.3	Hypothesis Tests	325
7.4	Bibliographic Notes	348
7.5	Problems	349
8	Linear Regression Models	353
8.1	Introduction	353
8.2	Normal Linear Model	359
8.3	Normal Distribution Theory	370
8.4	Least Squares and Robustness	374
8.5	Analysis of Variance	378
8.6	Model Checking	386
8.7	Model Building	397
8.8	Bibliographic Notes	409
8.9	Problems	409

9	Designed Experiments	417
9.1	Randomization	417
9.2	Some Standard Designs	426
9.3	Further Notions	439
9.4	Components of Variance	449
9.5	Bibliographic Notes	463
9.6	Problems	464
10	Nonlinear Regression Models	468
10.1	Introduction	468
10.2	Inference and Estimation	471
10.3	Generalized Linear Models	480
10.4	Proportion Data	487
10.5	Count Data	498
10.6	Overdispersion	511
10.7	Semiparametric Regression	518
10.8	Survival Data	540
10.9	Bibliographic Notes	554
10.10	Problems	555
11	Bayesian Models	565
11.1	Introduction	565
11.2	Inference	578
11.3	Bayesian Computation	596
11.4	Bayesian Hierarchical Models	619
11.5	Empirical Bayes Inference	627
11.6	Bibliographic Notes	637
11.7	Problems	639
12	Conditional and Marginal Inference	645
12.1	Ancillary Statistics	646
12.2	Marginal Likelihood	656
12.3	Conditional Inference	665
12.4	Modified Profile Likelihood	680
12.5	Bibliographic Notes	691
12.6	Problems	692

Appendix A. Practicals

696

Bibliography

699

Name Index

712

Example Index

716

Index

718